MIL-S-46777C
10 March 1981
SUPERSEDING
MIL-S-46777B
1 August 1973

MILITARY SPECIFICATION

SEMITRAILERS, DUMP, COMMERCIAL,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various sizes of rear dump semitrailers. The dump semitrailers are operated by the truck-tractor's hydraulic system, unless the self-contained engine driven hydraulic system option is required.

1.2 Classification. The semitrailers shall be of the following sizes, as specified (see 6.2):

Size 10	Min. 10-ton Payload, Refuse Special, 47 Cu. Yd.	MIL-S-46777/1
Size 12	Min. 12-ton Payload, Min. 8 Cu. Yd.	MIL-S-46777/2
Size 20	Min. 20-ton Payload, Min. 23 Cu. Yd.	MIL-S-46777/3
Size 25	Min. 25-ton Payload, Min. 40 Cu. Yd.	MIL-S-46777/4
Size 30	Min. 30-ton Payload, Min. 20 Cu. Yd.	MIL-S-46777/5
Size XX	As specified by procuring activity.	

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-514	- Plates, Identification, Instruction, and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-52895	- Chassis-Cab, Multi-Axle, On-Off Road, Commercial Construction Equipment (CCE). General Specification for.
MIL-T-52901	- Truck Tractor, On-Off Road Engineer, Min. 120,000 GCWR, Commercial Construction Equipment (CCE).
MIL-S-46777/1	- Semitrailer, Dump, Commercial Size 10, Refuse Special.
MIL-S-46777/2	- Semitrailer, Dump, Commercial Size 12.
MIL-S-46777/3	- Semitrailer, Dump, Commercial Size 20.
MIL-S-46777/4	- Semitrailer, Dump, Commercial Size 25.
MIL-S-46777/5	- Semitrailer, Hopper Bottom Dump, Commercial Size 30.

FEDERAL

KKK-T-2108	- Trucks and Truck Tractors; Commercial, Diesel and Gasoline Engine Driven, 24,000 to 32,000 Pounds GVW, 4 x 2.
KKK-T-2109	- Trucks and Truck Tractors; Commercial, Diesel and Gasoline Engine Driven, 34,500 to 64,000 Pounds GVW, 6 x 4.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-1223	- Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MS75021	- Connector, Receptable, Electrical – 12 Contact, lntervehicular, 28 Volt, Waterproof.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

M-931 - Highway Trailers, All Types, for TOFC Service.

(Application for copies should be addressed to the Association of American Railroads, 1920 L Street, NW, Washington, DC 20036.)

ALUMINUM ASSOCIATION (AA)

Aluminum Standards and Data

(Applications for copies should be addressed to the Aluminum Association, 818 Connecticut Ave., NW, Washington, DC 20006.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B93.31 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.
ANSI Z35.1 - Industrial Accident Prevention Signs, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 242 - High-Standard, Low-Alloy Structural Steel.
ASTM A 449 - Quenched and Tempered Steel Bolts and Studs.

(Application for copies should be addressed to the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welder Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN TRUCKING ASSOCIATIONS, INC. (ATA)

The Maintenance Council (TMC-ATA):
ATA Vehicle Maintenance Reporting Standards.
TMC-ATA Recommended Practice Manual (Formerly RCCC).

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(Applications for copies should be addressed to The Maintenance Council - ATA, 1616 P Street, NW, Washington, DC 20036.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure.
Welding Handbook.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

DEPARTMENT OF LABOR (DOL)

29CFR1910	- Occupational Safety and Health Standards. (OSHA).
29CFR1926	- Construction Safety Standards.

(Application for copies should be addressed to the local DOL Office or to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR325	- DOT, Compliance with Interstate Motor Carrier Noise Emission Standards.
49CFR393	- DOT Federal Motor Carrier Safety Regulations (FMCSR).
49CFR567 and 568	- Certification.
49CFR571	- DOT Federal Motor Vehicle Safety Standards (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

FORGING INDUSTRY ASSOCIATION (FIA)

The Forging Industry Handbook

(Applications for copies should be addressed to the Forging Industry Association, Dept. R, 1121 Illumination Building, Cleveland, OH 44113.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC. (FMS)

FMS Automotive Data Book.

(Application for copies may be obtained from commercial brake lining contractors or should be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Paramus, NJ 07652.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

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IFI - Standards
OMFS - Recommendations (Optimum Metric Fastener System).

(Application for copies should be addressed to Industrial Fasteners Institute, 1505 East Ohio Building, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW, Washington, DC 20005.)

NATIONAL WHEEL AND RIM ASSOCIATION (NWRA)

NWRA Standards.

(Applications for copies should be addressed to the National Wheel and Rim Association, 4836 Victor Street, Jacksonville, FL 32207.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE & RIM ASSOCIATION, INC. (T&RA)

T&RA Year Book.

(Application for copies should be addressed to the Tire & Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK BODY AND EQUIPMENT ASSOCIATION, INC. (TBEA)

Hydraulic Hoist and Dump Body Classification Chart.
Glossary of Terms; Hydraulic Hoists, Dump Bodies, Elevating Tailgates.

(Application for copies should be addressed to the Truck Body and Equipment Association, Inc., Suite 1220, 5330 Wisconsin Avenue, NW, Washington, DC 20015.)

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TRUCK TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA	- Recommended Practices
TTMA	- Technical Bulletins

(Application for copies should be addressed to the Truck Trailer Manufacturers Association, 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

3. REQUIREMENTS

3.1 Description. The dump semitrailer shall be as specified herein and on the applicable specification shall (see 1.2). The complete dumper vehicle shall include the following:

(1) Dump Body (see 3.4).
(2) Hydraulic hoist system (see 3.6 thru 3.8).
(3) Semitrailer components (see 3.9).
(4) Towing prime mover vehicle and components, when applicable (see 3.1.3).

The semitrailer, including attachments, equipment, and accessories, shall be the manufacturer's standard or optional items which meet or exceed the requirements as specified herein. Established component and vehicle ratings shall be manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. The semi-trailer shall be a commercial motor vehicle product, complying with all DOT Federal Motor Vehicle Safety Standards and DOT Federal Motor Carrier Safety Regulations applicable on date of manufacture to nonmilitary, nonexport motor vehicle products. The semitrailer vehicle identification number shall be in accordance with TTMA RP No. 56. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Unless otherwise specified in procurement documents, vehicle components shall be as represented and rated in the manufacturer's technical data, and mounted equipment and body shall be as represented and rated in the equipment and body manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of the procurement activity prior to delivery of the items (see 6.7).

3.1.1 Vehicle weights and dimensions. The dump semitrailer weights and dimensions shall be in accordance with table I and DOT 49CFR567 and 568, Certification. The ratings shall be in accordance with TTMA RP No. 39 and No. 53. The dimensions shall be in accordance with SAE J1100 and SAE J701, and as shown in figure 1. The curb weight shall include the weight of the semitrailer in operating condition with all attachments, equipment, and accessories, except for exclusion of the power-takeoff and other components intended for installation on the truck-tractor. The GVWR shall include the curb weight and the specified payload required in table I and the applicable specification sheet. The payload shall be evenly distributed over the dump body area. The physical characteristics of the semitrailer shall be as specified in table I, when the floor of the semitrailer body is in travel position on level ground, and the inflation pressure of the tires is as recommended in the TRA Year Book for GVWR operation at highway speed.

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TABLE I. Weights and Dimensions.

	Feature			Requirement
a.	Weights:			
	Mfgr's Rated GVWR			Per Specification Sheet.
	Payload			Per Specification Sheet.
	Axle Weight Limit			
	(a) Single Axle Rear Suspension			Max. 18,000-lb.
	(b) Multiple Axle Rear Suspension (each axle)			Max. 16,000-lb.
	Front Coupler Weight Limit			Max. 48% of GVWR.
	Rear Running Gear Weight Limit			Max. 68% of GVWR.
b.	Dimensions per SAE J1100:			
	Overall Length	(OAL)	L108	Per Specification Sheet.
	Overall Width	(OAW)	W103	Max. 98-in.
	Overall Height	(OAH)	H153	Max. 144-in.
	Ground Clearance	(GC)	H156	Per Specification Sheet.
	Angle of Departure Max., Degrees		H107	Per Specification Sheet.
c.	Interchange Dimensions per SAE J701 and J1100:			
	Front Support Clearance, Radius,	(LWC)	L498T-*	Min. 90-in. for Tandem Axle Tractor. Min. 72-in. for Single Axle Tractor.
	Swing Radius,	(SR)	L497T-*	Max. 58-in. (Except where specified in specification slash sheets.)
	Kingpin Setting,	(FP)	L496T-*	24 to 36-in. Range. (Except where specified in specification slash sheets)
	Level Coupler Plate Height	(FWH)	H494T-*	50 to 54-in. Range.
d.	Body Characteristics:			
	Struck Capacity, SAE J741			Per Specification Sheet.
	Heaped Capacity, SAE J741			Per Specification Sheet.
	Length of Body,		L598T*	Per Specification Sheet.
	TBEA Hoist Classification,			Per Specification Sheet.
	Body Overhang Beyond Rear Tires,		L495T*	Min. 18-in.

*Tentative and subject to change in SAE J1100.

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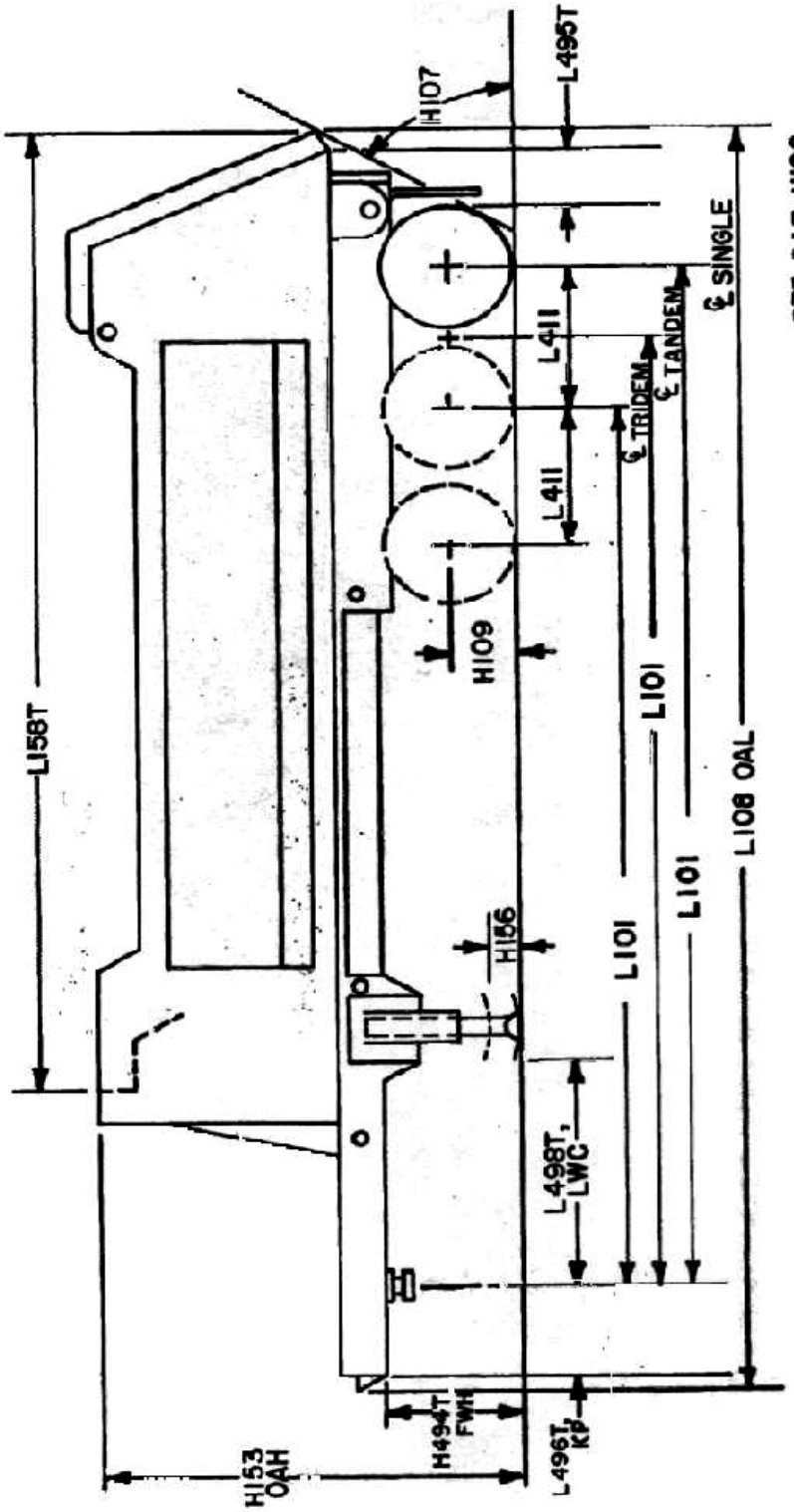


FIGURE 1. Dump semitrailer.

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3.1.2 Towing compatibility. Unless otherwise specified, the semitrailer shall be compatible for coupling, towing, maneuvering, spotting, and dumping under all conditions specified herein when coupled to a commercial truck-tractor that is unmodified, except for installation of power hoist components. Commercial truck-tractors shall include those specified in KKK-T-2108, KKK-T-2109, and MIL-C-52895, MIL-T-52901. The empty height of the underside of the upper fifth wheel plate above level ground, when the semitrailer is level, shall coincide with the height of the truck-tractor, having a rating equal to or exceeding the gross combination weight rating (GCWR) of the truck-tractor and semitrailer with payload. Each semitrailer shall be compatible with fore and aft rocking lower fifth wheels. When and as specified by the procuring activity (see 6.2, 4.5.2.5, and 6.4):

(a) The dump semitrailer shall be matched to the user's prime mover (truck-tractor) to insure complete, compatable, powered combination vehicle(s); or
(b) The dump semitrailer shall be furnished coupled to a new prime mover as a complete, compatable, powered combination vehicle package.

3.1.3 Material design criteria. The stresses in the load carrying members of the semitrailer, with the payload as specified in table I and the applicable specification sheet, shall not exceed 33 percent of the yield point (YP) strength under static conditions. A calculated stress analysis of the semitrailer shall be furnished. This analysis shall include calculations and diagrams for the following:

(a) The GVWR load distribution diagram of the semitrailer.
(b) The vertical shear diagram for the length of the semitrailer.
(c) The bending moment diagram for the length of the semitrailer.

3.1.4 First-article semitrailer. The contractor shall furnish a semitrailer of each size for examination and test within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce semitrailers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first-article examination and tests, as specified (see 6.2).

3.2 Special requirement. Semitrailers shall meet the following special requirements as specified herein.

3.2.1 Electromagnetic interference. When specified (see 6.2), semitrailers supplied with an auxiliary engine (see 3.8) shall have electronmagnetic interference emission and susceptibility characteristics conforming to SAE J551.

3.2.2 Treatment, painting, and data plates. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, and data plates shall be in accordance with MIL-STD-1223 and as specified herein. Inaccessible areas of the semitrailer, and truck shall be

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treated with anti-rust coating material in accordance with TTMA Technical Bulletin No. 56. When specified (see 6.2), the semitrailer shall be treated and painted in accordance with MIL-T-704, type A. Further, when specified (see 6.2), the multiple coat application requirement of MIL-T-704 may be reduced to a single coat having a minimum 10-mil dry thickness, when the semitrailers are painted with a color inpregnated, rust preventative coating of thixotrapic, solvent cutback, wax-based compounds having self-healing features.

3.2.2.1 Data plates. Data plates shall be applied to the semitrailer on metal plates in accordance with the "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers, Inc., (NAME). Identification and shipping data plates shall be of material in accordance with MIL-P-514, composition C. The plates shall be secured to each end item in accordance with the NAME Standards and Practices.

3.2.2.1.1 Identification. The semitrailer shall be identified in accordance with DOT, the manufacturers, and MIL-STD-130 on MIL-P-514, type I, style 1 plates. The date of the delivery shall be that of the complete vehicle(s) under contract and shall not be prior to date of Government acceptance.

3.2.2.1.2 Shipping data. The semitrailer shall have shipping data applied to MIL-P-514, type III plates. These plates shall show the silhouette of the end items, in transport position, indicating the empty center of gravity and the semitrailer manufacturer's approved points capable of being lifted or tied-down or, when furnished, the location and capacity of the lifting and the tiedown attachments. The plates shall be attached in a conspicuous protected location on the curbside.

3.2.2.1.3 Instruction. Each semitrailer shall be equipped with instruction plates, decals, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z35.1 and compatible with the space available.

3.2.2.1.4 Wiring diagram. The vehicular electrical wiring diagram plates or decals shall be furnished on the semitrailer, following TTMA Technical Bulletin No. 65.

3.2.2.1.5 Operating instruction. Operating instruction plates or decals shall be furnished and shall be permanently affixed in a readable location near the applicable components. Instructions shall be clear and concise and shall contain sufficient information to enable operation without damage to the components or injury to personnel.

3.2.3 Marking. As specified by the procuring activity for the appropriate service (see 6.2), marking and registration numbers shall be in accordance with MIL-STD-1223. The GVWR shall be marked on each forward front side of each body and the tire pressure (TP) shall be marked near the tires.

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3.2.4 Safety. The safety characteristics of the semitrailer shall meet the following minimum requirements:

(a) Minimize danger created by the equipment design.
(b) Minimize hazards to personnel and to property.
(c) Result in furnishing material having a correct safety factor.
(d) Provide safety and protective devices which do not impair the operating functional requirements or furnish prominent warnings.
(e) Provide correct safe guarding, enclosing, and insulating features for exposed components and systems which are subject to high temperatures, high pressures, electrical actuation, or which are inherently hazardous.
(f) Comply with SAE J185 on all walkway and step surfaces and handholds.
(g) Eliminate nonfunctional sharp edges, projecting points, and excessive length of fasteners.
(h) Incorporate human factors engineering characteristics into the semitrailer; provide the operating characteristics of the semitrailer for operation by personnel ranging from the small person, clothed, through the large person, arctic clothed, in accordance with SAE J833 and SAE J925.
(i) Conform to the California Vehicle Code, Division 12.
(j) Conform to DOT Federal Motor Vehicle Safety Standards (FMVSS).
(k) Conform to DOT Federal Motor Carrier Safety Regulations (FMCSR).
(l) Conform to the DOL OSHA Standards and the Construction Safety Standards applicable to for dump vehicles employed in construction work.

3.2.5 Lifting attachment. When specified (see 6.2), the semitrailer shall be provided with lifting attachments conforming to MIL-STD-209. The attachments shall enable the empty semitrailer to be lifted in its normal traveling or operating position. The attachments shall not result in damage or permanent deformation of the slings or of the exterior of the semitrailer. No spreader bars are permitted. Each attachment shall withstand not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load, when such limitation is possible. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall be not less than 3 inches. Members to which attachments are fastened and the fastenings themselves shall withstand all stresses imposed by the attachment. Loadings imposed when any two diagonally opposite lifting attachments and corresponding sling legs suspend the complete semitrailer shall not cause permanent deformation or racking damage. Where required structural members shall be reinforced to meet the lifting requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.2.6.

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3.2.6 Tiedown attachments. When specified (see 6.2), the semitrailer shall be provided with integral attachments conforming to MIL-STD-209 to permit tiedown of the semitrailer to the floor or deck of the transportation medium. The tiedown attachments shall withstand total loads of 8 G forward, 8 G rearward, 2-1/4 G upward, and 1-1/2 G sideward where G equals curb weight of the semitrailer (see 3.1.1). Each attachment shall be rounded or beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.2.5.

3.3 Performance. The dump semitrailer shall perform as specified herein when (1) equipped with all attachments, equipment, and accessories, (2) loaded to rated payload, not less than that specified in table I and the applicable specification sheet, uniformly distributed over the dump body load area, (3) completely serviced and adjust as specified in 3.12 and 3.13, and (4) towed over the roads and terrain as specified by a truck-tractor matched to the semitrailer GVWR and equal to he truck-tractors specified in 3.1.2.

3.3.1 Operational compatibility. The dump semtrailer shall operate with the following construction equipment:

(a) Concrete Pavers.
(b) Asphalt Paver.
(c) Chip Spreaders.
(d) Rock Crushers.
(e) Conveyors.
(f) Storage Bins and Hoppers, having a minimum underneath clearance of 144-inches above level ground.
(g) Asphalt Pug Mills.
(h) Scoop Loaders, 2-1/2 through 5 cubic yard size having bucket loading height of not less than 102-inches above level ground to lift and dump payload over-the-sides of the dump body, without side boards.
(i) Power Shovel, Clamshell, Dragline and Back Hoe up to 2 cubic yard size.

3.3.2 Payload handling. The dump semitrailer shall comply with the interrelated requirements as follows:

(a) Perform dumping cycles involving the transporting of payload materials in the function of loading, hauling, dumping, and returning empty.
(b) Dump in a fixed location, such as stockpiling.
(c) Dump in a controlled, uniform, spreading lift while traveling forward.
(d) Load, negotiate, and lump on off-road terrain, longitudinal grades, side slopes, and uneven ground normally encounted at earthmoving construction job sites.
(e) Except for the refuse specials, other semitrailers load, transport, and dump payloads of sand and gravel aggregates, 3-inch minus crushed rock, hot paving mixes, earth,

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clay, rubble and large boulders and shot rock having a largest diameter dimension of at least 5-feet.

(f) Perform short haul cycles on secondary roads of under 3-miles for at least 10 hours per day.
(g) Perform long haul cycles on secondary roads and sustained average high speeds of minimum 44 mph on highways for a minimum of 200 miles per day.
(h) While dumping, comply with DOT Noise Emission Standards (49 CFR 325) and DOL OSHA stationary requirements on noise level of the semitrailer and contractor furnished components.

3.3.3 Mobility characteristics. The dump semitrailer shall conform to the mobility characteristics specified herein without malfunction, permanent deformation, or damage. The semitrailer shall:

(a) Travel over smooth primary paved highways at minimum 55 mph speed.
(b) Travel over gravel and dirt surfaced secondary roads, haul roads, and trails at minimum 30 mph speed.
(c) Negotiate off-road terrain, grades, side slopes, mud, sand, and snow to and from operational sites.
(d) Negotiate hard bottom, creek water crossings at a minimum 20-inches deep without immersion damage.
(e) Negotiate highway V-ditches without damage (standard FHWA Bureau of Public Roads type, 36-inches wide by 18-inches deep).
(f) Withstand immobilization and tow-out operations.
(g) With single axle, provide a minimum of 6-inches suspension articulation.
(h) With multi-axles, provide a minimum of 8-inches suspension articulation between all combinations of wheels, front-to-rear as well as diagonal.
(i) Withstand truck tractor maneuvering resulting in pivot turning of the running gear in place, both clockwise and counter-clockwise and provide truck-tractor jack-knife clearance.
(j) Comply with DOT FMCSR off-tracking tolerance.
(k) Provide minimum braking performance in compliance with DOT FMVSS, DOT FMCSR and the California Vehicle Code.
(l) Provide braking system in accordance with TTMA RP No. 12 and RP No. 44, including DOT legally applicable anti-lock requirements;
(m) Provide parking brakes capable of holding the applied wheels locked on longitudinal grades up to 30 percents.

3.3.4 Rail transport. The semitrailer shall withstand all forces encountered in Trailer-on-Flat-Car (TOFC) rail transport service without damage or permanent deformaton. For purposes of rail transport, the cargo shall be at 50 percent of the payload specified in table I and the applicable Specification Sheets, uniformly distributed over the body floor area, and with the payload center of gravity at 36 inches above the semitrailer floor.

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3.3.5 Reliability and maintainability.

3.3.5.1 Mean-time-between-failure (MTBF). The mean-time-between-failure (MTBF) shall be at least 80 hours.

3.3.5.2 Maintainability. The ratio of the man-hours of maintenance to the hours of operation preformed shall not exceed 0.30.

3.4 Dump Semitrailer components. The dump semitrailer components shall be recommended for the rear dump semitrailer application by the component manufacturers and shall withstand the imposed static and dynamic loads under the operating conditions specified herein. At the option of the contractors, the semitrailer may be of the frameless or chassis-frame type. The chassis-frame may be fabricated of steel or aluminum. However, when and as specified with aluminum bodies or light weight semitrailers (see 3.5.(6) and 6.2), the type of dump semitrailer selected shall not exceed the maximum curb weight limit specified by the procuring activity. The dump semitrailer shall conform to all legal requirements and the maximum practical standardization requirements of table II.

TABLE II. Legal and Standardization.

Feature	Requirement
a. Legal Requirements -	(1) DOT Certification 49CFR567 and 568. (2) DOT FMVSS, 49CFR571. (3) DOT FMCSR, 49CFR325 & 393. (4) California Vehicle Code. (5) DOL OSHA, 29CFR 1910 & 1926.
b. Industry Standardization -	(1) Society of Automotive Engineers (SAE), including SAE Handbook. (2) Truck Trailer Mfgrs Association (TTMA), including TTMA RP's and TB's. (3) The Maintenance Council-American Trucking Association (TMC-ATA), including the TMC-ATA Recommended Practice Manual. (4) Truck Body and Equipment Association, including Classification of Hoists and Glossary of Terms. (5) Tire & Rim Association (TRA), including TRA Yearbook. (6) Friction Materials Standards Institute (FMSI), including FMS Automotive Data Book. (7) Industrial Fasteners Institute (IFI), including IFI Standards and OMFS Recommendations.

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3.5 Dump body. The semitrailer shall be equipped with a rear dump body from which the load is gravity dumped by opening the rear tailgate and raising the front end of the body, unless otherwise specified in the specification sheet. The minimum body volume struck and heaped capacities, as defined in SAE J741, and the dimensional limits shall be as specified in table I and the semitrailer specification sheet. Except as otherwise specified, the dump body shall be as follows:

(a) Unless otherwise specified by the procuring activity, provide dump body of high strength low alloy (HSLA) steel.
(b) When specified (see 6.2), provide dump body of aluminum alloy.
(c) Material used in steel body per ASTM A242 or SAE J410 HSLA steel having a minimum 50,000 psi guaranteed yield point (YP).
(d) Material used in aluminum body per "Aluminum Standards and Data" of the Aluminum Association and having a minimum 26,000 psi guaranteed YP.
(e) Floor of one piece of butt-welded metal plate(s), reinforced by cross members or interlaced understructure, or, for bathtub dumps, encircling transverse bows.
(f) Head sheet and sides inherently rigid or reinforced by shapes or members to withstand dumper service.
(g) Upper sides provided with box type side rails to absorb arm contact from scoop loaders while loading.
(h) Provide both front and rear body higher than the height over the sides.
(i) Provide standard 2-3/4-inch nominal size slots on both front and rear corners for accepting sideboards up to 12-inches high to be provided by the users.
(j) Provide horizontal members of the angles, dirt-shedder type to preclude dirt dropping from the semitrailer while traveling (not applicable to catwalks).
(k) Provide body access steps per SAE J185 on the left front exterior of the dump body.
(l) Provide a positive, hoist locking OSHA maintenance strut assembly to hold the dump body safely in the raised position in accordance with OSHA Regulations (29CFR 1926); in addition, provide strut assembly to hold at a point-selected between 1/2 and 3/4 of full dump body hoist arc angle; provide strut assembly which is self-alining to the holding device when the body is raised and then lowered.
(m) Provide a slanted, double acting tailgate having either cast or flame cut plate, off-set type, top hingers; provide both horizontal and vertical bracing of tailgate sheet; provide chains for both controlled payload spreading and for lowering the tailgate level with the bed of the body to accommodate extra long loads; provide minimum body overhand rearward of the tired for asphalt pavers of the rear-wheel roller types as specified in table I, d.
(n) Unless otherwise specified, provide manual tailgate control having lever on left front of body, a chained safety loop for securing lever, and nonbinding rods and mechanisms.
(o) When specified (see 6.2), provide an air operated tailgate trip control, operated by the driver from his seated position in the truck-tractor.

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(p) When specified (see 6.2), provide the availability of air pressure for the tailgate trip control only during the time that the driver presses down on the service brake air treadle in the cab, thus preventing unintensional tailgate release.

(q) Provide dump body load cover mechanism in compliance with all state laws pertaining to highway transport of loose materials.

(1) Provide a heavy weight elastomer plastic impregnated fabric tarpaulin intended for dump usage, sized to completely cover a heaped payload of earth and provide shock cord tie downs along the sides.

(2) Provide a tarpaulin which will not adhere to or be damaged by hot-mix asphalt up to 375° F.

(3) Provide the cover mechanism, mounted on the front of the semitrailer that, when actuated, will maintain the tarpaulin clear of the payload during covering and only near the end of travel permit the tarpaulin to be dragged over the aggregate.

(4) Provide driver controls for covering and for uncovering the load, of the manual, remote control types; location of controls for operation from the front, roadside, while the driver remains safely standing on the ground.

(5) Provide shock cords, preventing the air stream from lifting the tarpaulin and blowing loose materials, along at least every 36 inches on the side of the body; provide shock cord installation and removal method by the driver while safely standing on the ground.

3.6 Hoist system. The dump semitrailer shall be equipped with a front mount, telescopic ram, hydraulic system as specified herein and on the dump semitrailer specification sheet. The classification of the hoist system shall be in accordance with the Hoist Classification Chart of the Truck Body and Equipment Association (TBEA). The dump body shall hoist to the standard dump angle of the semitrailer manufacturer; however, unless otherwise specified, (see 6.2 and the applicable specification sheet), this angle shall be not less than 40 degrees. At least 6-1/2 inches of clearance shall be provided between hydraulic system components and lines and any heat radiating part of engine exhaust systems. The hoist components shall not bellmouth, bind, malfunction, or be damaged by the act of dumping the loaded semitrailer as specified herein, including dumping on side slopes up to the limit shown in figure 2. Stroke limiting hydraulic valve or other safe restraining devices shall be furnished to prevent overtravel of the hoist mechanism. Semitrailers with dump bodies that are 20 feet long and longer shall be equipped with either a hydraulic or cam stabilizing system or an air-spring suspension system which shall prevent rear suspension oscillation and decrease the risk of laterial instability during the dump cycle on firm ground.

3.6.1 Hoist actuation. Unless otherwise specified herein, the actuation of the body hoist system shall be as specified in (a) as follows. When specified (see 6.2 and 6.4), the actuation of the body hoist system shall be as specified in (b) as follows:

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(a) Truck-tractor hydraulic hose type, whereby hydraulic hoists pressure is provided from the towing truck-tractor's hydraulic system (see 3.7.1(a) thru (h)).
(b) Self-contained type, whereby the complete hoist system is mounted on the semitrailer with a gasoline or diesel engine (see 3.8) driving the hydraulic pump (see 3.7.1 (a) and (c) thru (h)).

3.7 Hydraulic system. The dump body shall be actuated by a front mounted telescopic ram hydraulic hoist system having a minimum class number specified in the specification sheet's TBEA Hoist Classification, and a minimum ton rated capacity as specified in the specification sheet. The hydraulic hoist system shall be as follows:

(a) Provide the complete hydraulic system having all components as recommended by the hydraulic pump hoist cylinder manufacturer's recommendations:

(1) Operate with rated payload without binding, malfunction, or permanent deformation.
(2) Provide a fail-safe system protected from accidental lowering due to any hydraulic system failure other than telescopic cylinder.
(3) Capable of withstanding minimum proof pressure without leakage as recommended by the pump manufacturers.
(4) All components and lines cleaned and free of foreign matter.
(5) Hydraulic oil approved by the pump mfgr. for severe heavy-duty service.

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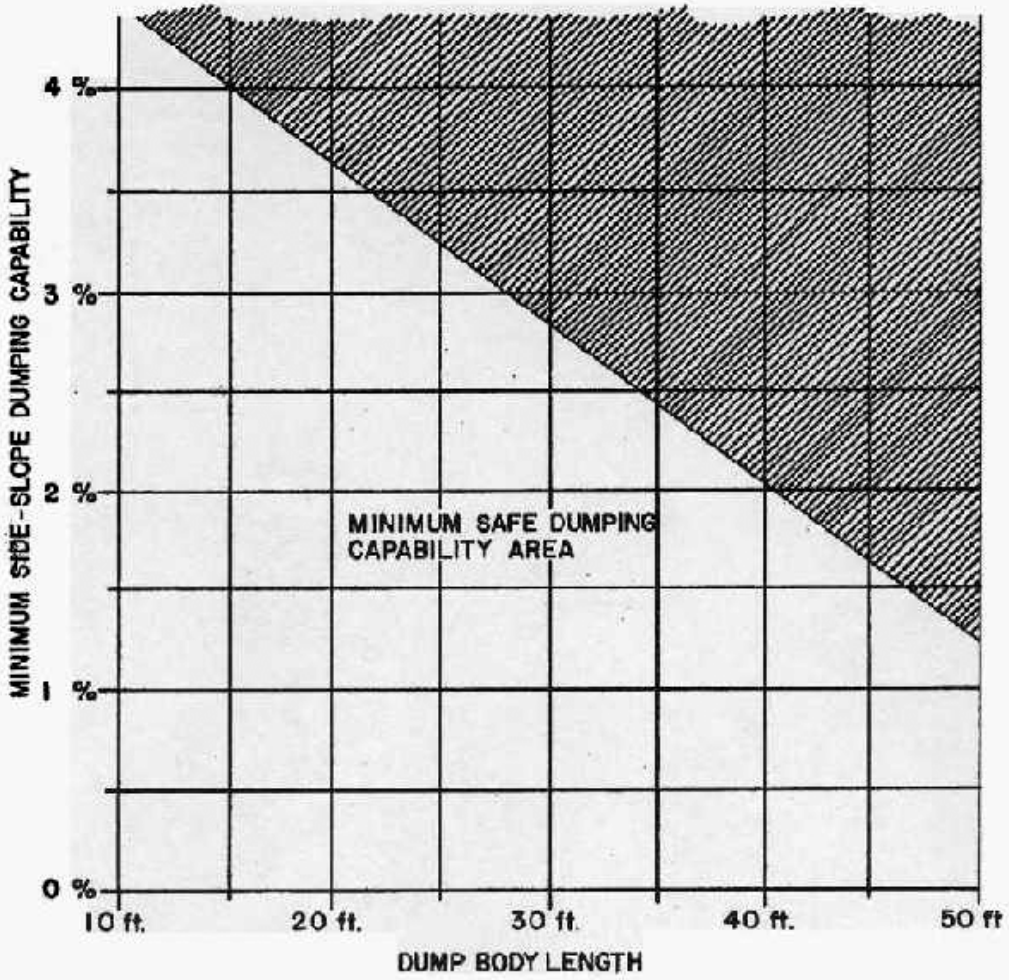


FIGURE 2. Dump semitrailor minimum side slope operation.

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(b) Provide power-take-off (PTO) in accordance with SAE J705 and J772 of the antifriction bearing, single speed, hydraulic clutch pack "hot-shift" type having an intermittent service rating of at least 120 percent of the maximum needs of the driven components. Provide means to prevent overspeed of the PTO and its driven components.

(c) Provide a hydraulic hoist pump in accordance with SAE J744 of the positive displacement type:

(1) Provide pump integrally mounted on the power-take-off (PTO) on the main transmission or, when self-contained, integrally mounted on the engine.
(2) Provide a pump having the pump manufacturers's ratings for speed and performance equal to or in excess of the required speed and performance necessary for dumper hoist operation.
(3) At engine rpm recommended by the pump manufacturer and at the specified pounds of payload, provide dump body hoist or extraction time of not more than 2-minutes and dump body lower or retraction time of not more than 2-minutes.
(4) Provide relief valve per SAE J747.
(5) Provide a capped pressure gage test point at the outlet of the pump.

(d) Provide a hydraulic reservoir as follows:

(1) Capacity matched to telescopic cylinder.
(2) Provide filler cap captive chained to a removeable, 60 mesh strainer in the filler neck, and marking "HYD RES" on or near reservoir cap.
(3) Provide self-sealing dipstick or sealed and guarded oil sight level indicator.
(4) Provide reservoir air breather(s) of the spin-on pleated paper canister type having a minimum size of 5-inches long by 3.7-inches diameter threaded on a SAE J363 3/4-16 UNF2A or 2B, insert.
(5) Provide reservoir mounting to prevent damage from vibration and damage from off-road terrain travel.

(e) Provide hydraulic system return oil filter in accordance with ANSI B93.31, as certified by the filter manufacturer, having the following:

(1) Minimum 6.2 filtration ratio at 20 micrometers and a dust capacity of 1 gm per gpm flow.
(2) Provide minimum 100 psi housing burst pressure.
(3) Provide automatic by-pass valve permitting full flow at 35 psid and greater.
(4) Provide a filter element service indicator gage.

(f) Provide a dump hoist control valve per SAE J747:

(1) Provide lever control, accessible to the driver from his seated position in the cab, or from his standing position on the ground at the front roadside of the semitrailer.
(2) Provide lever with tower type neutral "HOLD" lock of the hinged gate type, preventing accidental raising or lowering of the dump body.

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(3) Provide a lever to valve, flexible, push-pull control cable of the water-proof, noncorrosive and stainless steel SAE J917 type.
(4) Provide control valve of the multiposition, spring centered spool type having wiper seals.
(5) Provide dump operator feedback warning when dump body is nearing maximum dump angle.
(6) Provide hoist control identification and position labeling on quadrant with illumination for night operation and provide an on-off switch that is powered from the marker lamp circuit.

(g) Provide flexible hydraulic hose lines between all components as follows:

(1) Provide hose per SAE J516 and SAE J517 having a working pressure rating in excess of the relief valve setting.
(2) Maintain hose bend sharpness in accordance with SAE J517 limits.
(3) Provide fittings at ports of the components in accordance with SAE J514, O-rings boss type, or SAE J518.
(4) Provide test point connection at pump outlet line for measuring pressure.
(5) Provide lines with reusable fittings; provide replacement lines with reusable fittings and bulk hose size and length dimensions in the parts manual.
(6) When applicable, wet line from truck-tractor to semitrailer furnished with a minimum 120-in. length and with quick-disconnect fittings on semitrailer end (and, when included, on the truck-tractor end).

(h) Provide multi-segment telescopic hoist cylinder matched to the dump body having sufficient mounting structure to spread the loadings over the dump body framing and over the fifth-wheel or the chassis framing:

(1) Provide cylinder of seamless or welded and redrawn steel tubing having concentric bearing surfaces of the interstages.
(2) Provide hard chrome plating of cylinder exterior working surfaces to prevent corrosion
(3) Provide hydraulic cylinder fail-safe check valve or safety velocity fuse to lock the cylinder when oil escapes from the cylinder at a faster rate than normal design.
(4) Provide automatic means to bleed air from the hydraulic system and especially the hydraulic cylinder.
(5) Provide multi-stage telescopic cylinder wiping seals of the polyester elastomer type having inner and outer lips on each stage to restrict the entrance of dust and dirt ahead (outside) of the cylinder packings; wiping seal glands to fit NFPA wiper groove dimensions.
(6) Provide cylinder hydraulics which have a maximum down drift of 1-inch in a 30 minute time period when at rated payload and measured not more than 36-inches from the front of the dump body.

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3.8 Auxiliary engine. When and as the self-contained hoist system is specified in 3.6.1 (b), a gasoline or diesel engine shall be provided to drive the hydraulic hoist pump. Engine shall be a 4-cycle heavy-duty type and shall have sufficient output power to drive all operating components other than the vehicle. During operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein.

3.8.1 Engine components. The engine shall be a 4-cycle engine equipped with at least the following components:

(a) Solid state type ignition (gasoline only).
(b) Self-starter, 12 volt.
(c) Resistor-type spark plugs (gasoline only).
(d) Dry type, replaceable element air cleaner.
(e) Automatic choke (gasoline only).
(f) Tachometer.
(g) Ether cold weather starting system (diesel only).
(h) Air or liquid cooling system.

3.8.2 Engine control panel. The engine hand throttle, key-operated ignition switch, and palm starter button shall be installed in a weather-proof cabinet on a control panel, and each shall be identified as to their function. The cabinet shall be provided with an anti-vandal, hinged cover with a spring-type latch and provisions for locking with a padlock having a 3/8-inch shackle.

3.8.3 Engine electrical system. The engine electrical system shall be in accordance with DOT Federal Motor Carrier Safety Regulations 393.30 through 393.33. The battery furnished shall be a minimum size 30 or 31, 12-volt battery conforming to SAE J537. The battery shall be secured with battery hold downs in an enclosed battery box.

3.8.4 Engine fuel pump. A fuel pump shall be furnished and when lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line primary water separator and a secondary fuel filter shall be provided.

3.8.5 Governor. A variable-speed governor is required to be furnished. The governor shall automatically control the engine speed at the selected rated speed at all pressures required. A manually operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum idle speed up to the maximum governed speed.

3.8.6 Engine fuel tank. The engine shall be furnished with an engine fuel tank. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be painted red and shall be attached to the tank by means of a corrosion-resistant chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A

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drain shall be so located at the bottom of the sump for complete drainage of the fuel tank. Drainage shall be provided to preclude spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the line to the power source and shall be adjacent to the fuel tank. An identification plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap (see 3.2.2.1.5). Terne-plate fuel tanks shall not be supplied when diesel engines are furnished.

3.8.7 Exhaust system. The exhaust system shall be furnished with a legally quiet, corrosion-resistant muffler and means shall be provided to vent exhaust fumes away from operating personnel. Muffler and exhaust pipe shall be enclosed with a corrosion-resistant, ventilated metal shield. The outlet shall be provided with an elbow and counterbalanced rain cap to preclude weather entering the exhaust system. The rain cap shall open upon exhaust pressure.

3.9 Semitrailer. The semitrailer specifications shall be in accordance with table III. The component sequence and number system shall be in accord with the "Vehicle Maintenance Reporting Standards" (VMRS), the trucking industry standardization system of the American Trucking Associations (ATA).

TABLE III. Semitrailer specifications.

VMRS No.	Feature	Requirements
Group 10	Chassis. Provide strength, flexibility, and rigidity, to withstand dynamic transport, both highway and off-road, and to withstand loading and unloading operations.	
VMRS 12-001	Axle – Rear, Non-Driven. Provide single, 2-ea., or 3-ea., as specified herein, and in the applicable dump semitrailer specification sheet; Axles per DOT FMVSS and TTMA RP No. 39; Min. ratings per each axle as follows: - Min. 20,000 lb. partial rated GAW at hub for subsequent total GAWR at ground. - Min. 25,000 lb. bearing and axle beam rating. - Min. 20,000 lb. brake rating. Provide industry standarized 71.5-in. treat for disc wheel applications.	
VMRS 12-001-001	Tube – Rear Axle, Non-Driven. Hardened and ground steel spindles.	

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TABLE III. Semitrailer specifications (continued).

VMRS	Feature	Requirements
VMRS 13	Brakes. Per DOT FMVSS, DOT FMCSR, and California Vehicle Code. Straight air service and emergency-parking system. System withstand severe service including high speed, off-road, and mountainous travel and including creek bed immersion usage. Provide priority to brake system over any auxiliary air system use.	
VMRS 13-001-026	Disc – Brakes. High heat sink type having cooling features.	
VMRS 13-002	Rear Brakes & Drums. Disc type or wedge or S-cam drum type. Components protected from damage. All components located above the bottom of the brake disc or brake drum. Discs, wedges, cams and roller pins, as a minimum, phosphate coated, stainless steel, or hard chrome plated and having silicone high temp compound on brake bearing surfaces (except wedges and S-cams and rollers) to prevent corrosion malfunction. Provide drum forward rotation and S-cam rotation in the identical direction on all axles.	
VMRS 13-002-004	Anchor-Rear Brake Shoe. Drum brakes of single or double anchor type providing ease of maintenance features.	
VMRS 13-002-014	Lining – Rear Brake. Lining per SAE J866, Code EE, EF, or FF, having identical friction value on all lining segments; Lining selected from the FMS Automotive Data Book. Lining rated per DOT procedures and technically for the logging vocation or mountainous duty service. Wet lining fade not less than 65-percent of the dry friction baseline value, per SHE J786 test code.	
VMRS-13-002-023	Drum – Rear Brake. Drum brakes of outboard mounted type providing brake maintenance capability without hub removal.	

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TABLE III. Semitrailer specification (continued).

VMRS No.	Feature	Requirements
VMRS 13-002-028	Adjuster - Rear Brake Slack. Self-adjusters for each wheel providing correct compensation for lining wear without lock-up or unbalance and manual means to adjust the foundation brakes initially and in case of malfunction.	
VMRS 13-004	Spring Parking Brake. Provide 2 ea. (with single axle) or 4 ea. (with tandem and tridem axles) heavy duty spring type emergency-parking brake actuators on each axle, as necessary to prevent wheel rotation on grades up to a min. 30 %. Main springs of the elastomer coated type to restrict corrosion. Provide means on each semitrailer to cage the springs. Provide environmental protective covers, boots, and seals on each chamber.	
VMRS 13-010-003	Line - Air (Non - Metallic). All-hose type per TMC-ATA RP-305, having reuseable swivel fittings permitting hose replacement from bulk hose supply. Plumbing routed within the protection of framing members. Elastomer cushioned clamps and grommets or thru-frame connectors to secure and support hose. Support lines at a max. of 36-in. span. Min. 3-in. of hose slack at relay valves ports and identify each hose's function adjacent to each fitting with waterproof means (tags).	
VMRS 13-010-004	Tank - Air. Air reservoirs per DOT and SAE J10. Min. 2-ea. furnished, 1-ea. service, and 1-ea. Emer-Pk spring release, and (when furnished) l-ea. Air Suspension with isolating pressure protection valve. All reservoirs furnished with interior antirust coating. No component lower than bottom of the frame rails.	
VMRS 13-010-007	Valve - Ejector, Air Tank Moisture. Provide automatic drain valves on each air tank.	
VMRS 13-010-009	Valve - Relay, Air Brake. High speed type having waterproof features, including providing an exhaust hose long enough to preclude water entering relay valve during creekbed fording.	

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TABLE III. Semitrailer specification (continued).

VMRS No.	Feature	Requirements
VMRS 13-010-048	Valve - Release, Spring Brake. High speed type providing selective reservoir changing, spring brake parking application emergency application and anti-compounding features.	
VMRS 13-010-030	Air Line Coupler - Service and Emergency Gladhands per SAE J318, color coded yellow for control (service) on curbside, and color coded red for supply (emergency) on roadside. ID tags and captive chained manual gladhand dust covers. Spring load gladhand covers to prevent falling below the upper coupler plate. Gladhand location, 2-ea. on front of semitrailer per TMC-ATA RP-703 and SAE-J702	
VMRS 14-001	Frame Assembly. All welded construction having steel framing members or aluminum framing members. Frame type chassis furnished with the main frame longitudinal members of the continuous type. Frameless type chassis furnished with articulating elevating arms locating the kingpin and running gear under the body within industry tolerances.	
VMRS 16	Suspension. Unless otherwise specified, provide flexibly sprung, suspension of the equalizing type having min. 22,000 lb. per axle rating. When specified (see 6.2), provide an air suspension system having min. 20,000 lb. per axle rating. Provide single (1-axle), tandem (2-axle), or tandem (3-axle) as specified herein and in the applicable dump semitrailer specification sheet. Provide stabilized ride for loaded semitrailers with a high center of gravity without hazardous handling and provide soft, non-damaging ride for empty semitrailers. Provide suspension to protect semitrailer from damaging shock and vibration. Multi-axle suspensions having min. 8-in. of both front to rear and diagonal walk axle articulation. Installation and ajustment of suspension per both axle and suspension mfgrs. recommendations.	
VMRS 16-008	Air Ride. When specified (see 6.2) provide air-spring type intended for on-off road use.	

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TABLE III. Semitrailer specifications (continued).

VMRS No.	Feature	Requirements
		Min. 8-in. travel range from compression to rebound each axle. Cushioning bumper stops, anti-sway components linking road-side and curbside, and heavy duty shock absorbers dampening ride. Provide positive cushioning stops other than shock absorbers to halt rebound and withstand min. 2-g unsprung weight deceleration, typical of lowering the semitrailer with a crane. Air spring suspension of trailing member type, having piston or multiple bellows per SAE J551, internal bumpers. Provide air pressure protection valve, air controls, and air-tight plumbing. Plumbing of flexible hose and reusable swivel fittings, same as for air braking system. Ride frequency max. 130-CPM at center of dump body. Height control valves, closed center time delay type protected from road splash, debris, and obstacle damage. Time delay limits min. 3-sec. to max. 8-sec. Guarded driver over-ride air control on roadside front for selection of (1) high, maximum suspension height, (2) average, normal automatic height, and (3) low, depleted-air height.
VMRS 17-001-001		Tire – New. Tires comply with TRA Year Book and DOT FMVSS for highway speeds and 50 mph for ML tires. Tires at least 100 level quality intended for use on powered units. Provide all tires with traction-road-lug type tread. Unless otherwise specified, provide 2-ea. widebase single tubeless tires on each axle, min. size 18-22.5 ML, min. LR-H (16 PR). When specified (see 6.2), furnish 4-ea. conventional radial dual tubeless tires on each axle, min. size 12R22.5 min. LR-G (14PR). When specified (see 6.2), furnish 4-ea. conventional bias dual tube-type tires on each axle, min. size 11.00-22 min. LR-G (14PR); provide heat resistant radial type inner tubes, tire flaps, and multi-piece rims with the wheels. When specified (see 6.2), furnish 1-ea. inflated spare tire and wheel assembly with each semitrailer.

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TABLE III. Semitrailer specifications (continued).

VMRS No.	Feature	Requirements
VMRS 17-008	Valve. Provide metal valve caps w/elastomer seals for all tire assemblies.	
VMRS 18-003-001	Wheel Assembly Disc. Disc wheels per DOT FMVSS and TRA Year Book. Tire and wheel assemblies inherently in-balance. Disc wheels per SAE J694 10-hole heavy duty on-off road type having standard industry off-set. Wheels not to exceed max. 050-in. radial and laterial run-out (TIR) or Max. 120-in. oz. out of balance, to prevent wheel tramp or drumming. Wide base single type rim and disc made by the same mfgr. or fabrication approved by rim mfgr. For 18-22.5 tires, furnish size 22.5-14.00 wheels. For 12R22.5 tires, furnish NWRA Wheel No. X-4932 size 22.5-8.25 wheels. For 11.00-22 tires, furnish NWRA Wheel No. X-4927 size 22-8.0 wheels.	
VMRS 18-004	Hub, Rear-Wheel. Axle hubs per SAE J694, 10-stud, 11.25-in. BC, for disc wheels and with out-board mounted brake drums, when drum type brakes are furnished.	
VMRS 18-004-002	Bearing, Rear Wheel. Standardized tapered roller bearings with cups, bearings, and cones, all matched in accordance with the bearing and axle manufacturer's recommendations. Bearings per TMC-ATA Advisory AV6-15. Standardized bearing sets selected from the following:	
	Position	Prefered Set / Optional Set
	- Bearing cone, inner	HM218248 / JE9549
	- Bearing cone, outer	HM212049 / JHM516849
	- Bearing cup, inner	HM218210 / JE9510
	- Bearing cup, outer	HM212011 / JHM516810
VMRS 18-004-011	Seal - Oil, Wheel. Lube oil seals per TTMA RP No. 43.	
VMRS 18-004-017	Dust Cap, Rear Wheel Hub. Visual dust caps for oil lube level.	
Group 30.	Electrical Vehicle electrical system per DOT, California Vehicle Code, SAE Handbook, TMC-ATA RP Manual, and TTMA RP No. 9 and No. 14.	

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TABLE III. Semitrailer specification (continued).

VMRS No.	Feature	Requirements
	Heavy duty waterproof system providing protection from environment, physical damage, shock and vibration; 12-volt system potential with negative ground per SAE J538.	
VMRS 34-002	Tail and Stop Lamp. Furnish total of 4 ea. on rear of semitrailer providing functions of 2-ea. outboard turn and clearance plus 2-ea. inboard stop and tail. Lamps per TTMA RP No. 14 and SAE J759, coding SAE-IST, all of the flush-mounted type either elastomer ring with hermetrically sealed unit or elastomer housing with shock mount bulb socket, a snap-on or screw-on lens, and elastomer gasket. Round, 4-in. dia. size lamp installed flush in the TTMA standard 4.5-in. hole. Double filament configuration having 3-wire pigtail including a ground wire to the semitrailer framing member. Install lamps in rear frame in sub-frame cross member, or within body mounted boxes. Provide all lamps recessed forward from the body dump edge to aford lamp protection from backing into stockpiles and into loading docks.	
VMRS 34-002-007	Relfectors – Safety. Furnish reflectors per TMC-ATA RP-702 and TTMA RP No. 9 plus 2-ea. additional on the front facing forward. Secure all reflectors to the vehicle with metal fasteners. No reflectors installed on the tailgate because of shock breakage.	
VMRS 34-004-021	Lamps – Marker. All running lamps (clearance not in combination lamps, identification, and side marker) per SAE J592 and TMC-ATA RP-108, either flush mount type per RP-108/1 or in-line protective lens type per RP-108/3. All lamps equipped with polycarbonate plastic lenses and either hermetrically sealed shock resistant filaments or fleet service bulbs per TMC-ATA Advisory AY 1-2 having the trade number cited therein. All running lamps, clearance, identification, and side marker of same interchangeable make and model. Locations per TMC-ATA RP-702 and TTMA RP No. 9.	

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TABLE III. Semitrailer specifications (continued).

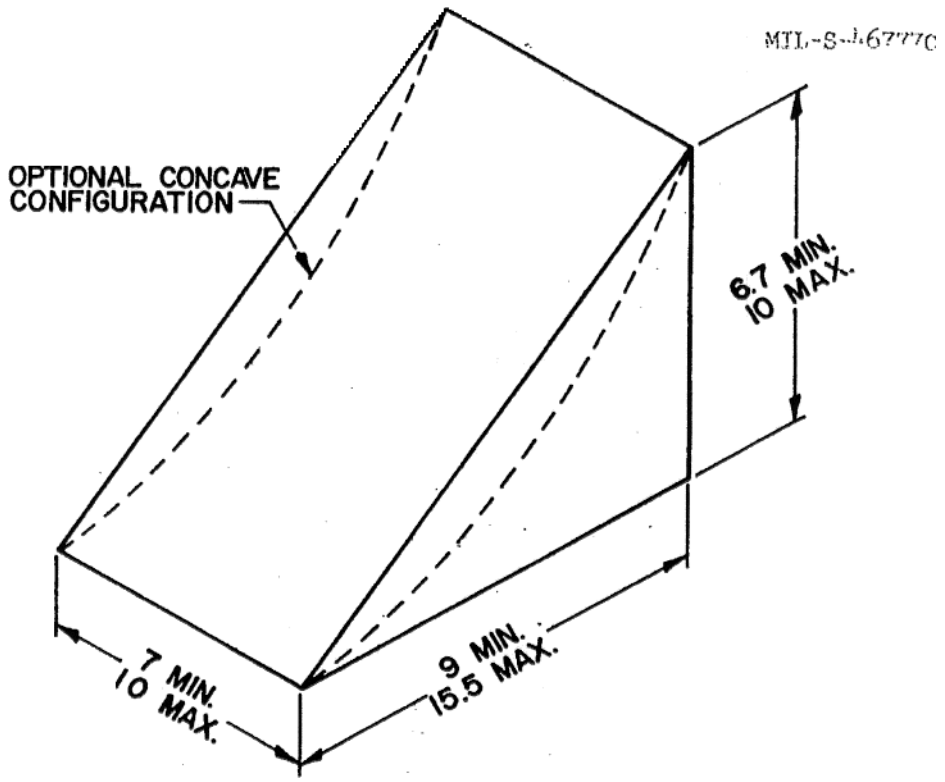
VMRS No.	Feature	Requirements
VMRS 34-003-032	Receptacle W/Circuit Breakers – Trailers.	
	Front 7-way receptacle installation per TMC-ATA RP-703 and similar to TTMA RP No. 6.	
	Receptacle per TMC-ATA RP-107, and TTMA RP No. 40 having min. 5 degrees water-drainage-slant and an elastomer truck-tractor plug support bracket or an extended lower lip providing support for the plug. Provide weatherproof connection box with nut and stud terminals.	
VMRS 34-004-010	Circuit Breakers.	
	SAE J553 circuit breakers protecting all live circuits leading from 7-way front trailer receptacle.	
	No fuses.	
VMRS 34-004	Wiring.	
	All wiring per SAE J163, SAE J555, TTMA RP No. 14, and the TMC-ATA RP Manual.	
	Wire identification codes per SAE J560 or, optionally, SAE J821.	
	Multiple circuits combined into harnesses.	
	Wiring supported at max. 36-in. spacing with anchored elastomer ties or coated clamps to prevent whipping, chafing, and mechanical or fatigue damage.	
	Interior of mechanical splicing devices waterproofed with RTV silicone sealant.	
	Wire per TMC-ATA RP-110 equipped with terminals per TMC-ATA RP 112.	
	Disconnect points for multiple circuits (in excess of 4 ea.) of the waterproof plug type or of the nut and stud junction block type.	
	Junction blocks coated with electrical varnish after assembly.	
	Wire-to-wire terminals secured to vehicle.	
	For positive return ground, provide ground wires between the front receptacle and the dump body and also between the front receptacle and chassis or the front upper couple plus the rear suspension framing members.	
	On a metal plate or decal (see 3.2.2.1.4) provide a wiring diagram, as actually furnished on the semitrailer, per TTMA Technical Bulletin No. 65.	

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TABLE III. Semitrailer specifications (continued).

VMRS No.	Feature	Requirements
Group 50	Accessories.	
VMRS 51-002-001	Hubodometer Assembly.	Furnish hubodometer of the driveless, tamperproof, self-lubricating type for converting wheel revolutions into accurate mileage readings. Visual recording counter having mileage numbers in the upright position. Min. 7-digit type recording a min. 999,999.9-mi. Corrosion-resistant housing sealed against moisture, oil, grease, dust, and climatic conditions. Hubodometer located on rear most curbside axle hub.
VMRS 53-004-013	Chock – Wheel	When specified (see 6.2), furnish 4-ea. Wheel chock blocks and either hangers or cradles, two assemblies on each side near the forward rear axle(s). Chocks blocks per Figure 3 and, when selected, cradle per figure 4. Captive chains of a length to place blocks both ahead of and behind the forward tires of the front axle without being taunt. Hangers of the latched securment type under the dump body. Cradles installed on a vertical flat surface with the ears up and the lower loop down having excess chain figure-eight-wound over the ears.

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT 11.5 LB.

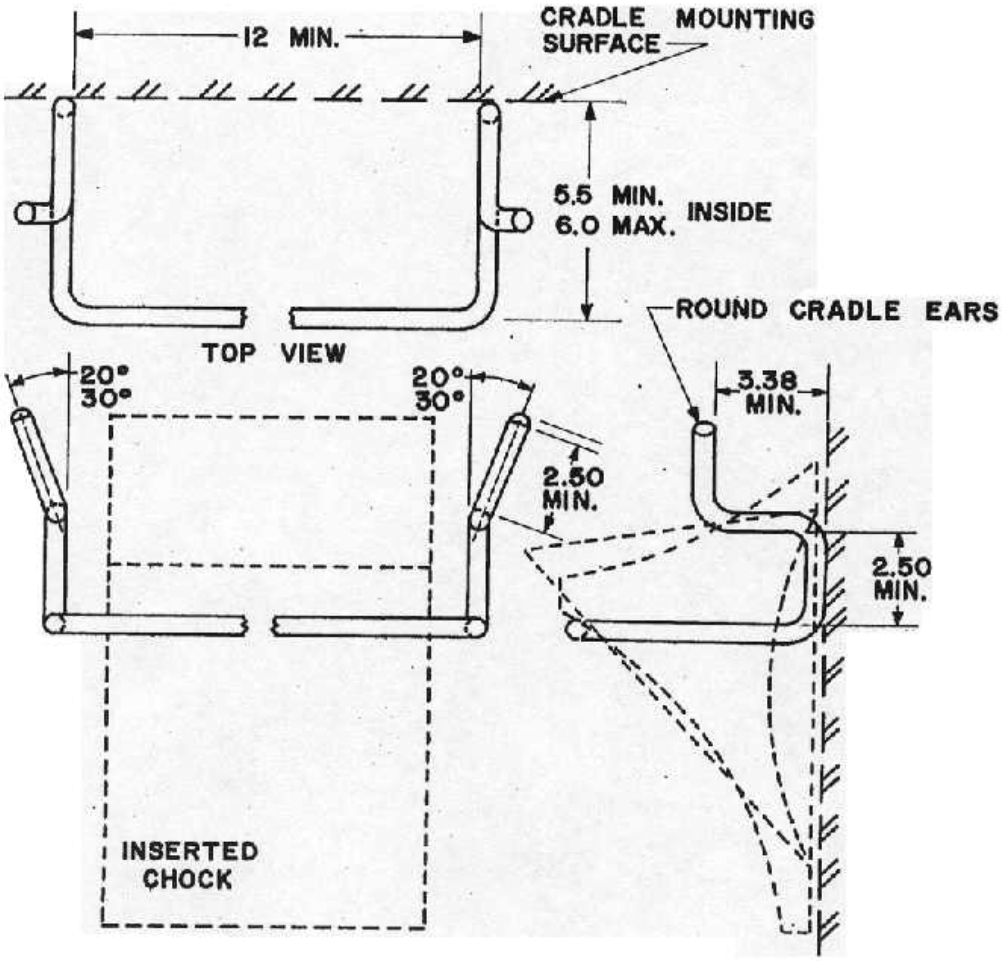
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 3. Chock block assembly.

X 630 A

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CRADLE MATERIAL NOTE: MINIMUM 1/2-INCH DIA. STEEL ROD. BEND RADII - 1.0-IN. MIN.

FIGURE 4. Cradle.

X-604A

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TABLE III. Semitrailer specifications (continued).

VMRS No.	Feature	Requirements
VMRS 77-002	Vertical Support.	Unless otherwise specified, furnish front vertical supports (semitrailer landing gear) per TTMA RP No. 4. Twin vertical lift, telescopic, non-rotating type supports with two speed gearbox and folding handcrank equipped with holder. Leg travel min. 17-in., providing upper coupler-plate- to-ground clearance range of at least 46-in. thru 58-in. at the kingpin centerline. Full skirt type support legs. Provide longitudinal bracing and transverse bracing. Support withstand loaded, free-fall drop of at least 4-in. height without permanent deformation. Gearboxes filled with reopectate silicone base grease per TMC-ATA Advisory AV 7-5. When specified (see 6.2), furnish front supports of the adjustable stiff-leg type having captive chained securing lock-pins and manufacturer's standard sand shoes.
VMRS 77-003-004	Sand shoe.	Provide equalizing sand shoes on each leg having turned-up edges. Min. 120-sq. inch area for each shoe.
VMRS 77-005	Upper Coupler Assembly	Upper coupler assembly per DOT FMCSR 393.70, TTMA RP No. 6, and No. 19, TOFC rated.
VMRS 77-005-001	Kingpin.	Kingpin of 2-in. dia. throat per SAE J700, SAE J133, and AAR Spec. M-931. Kingpin of alloy steel having either flame hardening or induction hardening types of heat treatment in the jaw wearing contact area of throat only.
VMRS 77-005-002	Plate – Top.	Front coupling ramp, min. 2-in. vertical height, fabricated of slit pipe or beveled plate with gussets and cover, no standing water traps. Coupler top plate of ASTM A242 or SAE J410 HSLA steel for both steel and aluminum dump bodies. Size min. 42-in. wide and completly cover a 36-in. dia., lower 5th wheel. Coupler top plate min. 1/4-in. thick. Plate and kingpin reinforcement of egg-crate type stiffener.

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TABLE III. Semitrailer specifications (continued).

VMRS No.	Feature	Requirements
		External SAE J534 lube fittings with lines to lubricate both road and curb sides of plate while coupled to the truck tractor.
VMRS 78-009	Deflector Assembly – Gravel	Splash guards per SAE J682, SAE J683, TTMA RP No. 41, and TMC-ATA RP-701. Provide rigid front guards and flexible top-side and rear guards for the trailer's running gear. Rear splash guards of the anti-sail type, min. size 30-in. long x 24-in. wide by l/4-in. thickness. Provide splash guards of reinforced, flexible elastomer having min. 3000 psi. ultimate strength and inherent cold weather resistance. Either provide splash guards having a textured splash surface of short blades of elastomer or mount and angle the brackets holding the guards inward approximately 5 to l0-degrees to reduce splash and aim the water-mist inward during rain.
VMRS 78-009-003	Bracket – Gravel Deflector.	Provide semitrailer brackets on running gear frame or on the dump body. Splash guards mounted to brackets by metal clamping straps fastened with hexbolts and nuts thru the TTMA/TMC-ATA standard hole pattern. Provide either captive chained alligator clamps or splash guard holder brackets for holding the guards raised and out of the way while the semitrailer dumps hot mix into asphalt paving machines of the type that drive thru the rear semitrailer's tires (all except refuse specials)

3.10 Military vehicle electrical system. When specified (see 6.2), the semitrailer shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact receptacle, the trailer shall be equipped with a military 24-volt, 12-contact intervehicle receptacle in accordance with MS75021-1, having a spring-loaded, waterproof cover. The military receptacle shall be located near the TTMA receptacle and both shall be labeled with identification plates (see 3.2.2.1.3) for 24 volts and 12 volts respectively, negative ground. The military 24 volt wiring shall be routed into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed water-tight. The military 24 volt circuits shall be connected through replaceable solid-state devices or resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with the following tabulation:

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Military-Commercial Connections

Function	Mil. Receptacle Terminal (Circuit No.)	Connection Instructions	Comm. TTMA Wiring Ident
Blackout Marker	A	None.	–
Ser. Left Turn & Stop	B (22 – 461)	Connect to –	YEL (3)
Blackout Marker	C	None.	–
Ground	D (90)	Connect to –	WHT (1)
Ser. Tail & Marker	E (21 – 489)	Connect to - & Connect to –	BLK (2) BRN (6)
Blackout Stop	F	None.	–
Blackout Marker	H	None.	–
Ser. Right Turn & Stop	J (22 – 460)	Connect to –	GRN (5)
Aux. Power (Hot)	K (37)	None	BLU (7)
Ground	L (90)	Ground to Frame	–
Electric Brake	M	None.	–
Electric Brake	N	None.	–

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits, letter coded B (461) and J (460) on the tactical trucks, are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military truck's 24-volt system input.

3.11 Fungus and moisture resistance. The electrical circuitry of all electrical systems, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with commercial electrical varnish:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.12 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication. The types of lubricants shall be in accordance with and selected from SAE J754 and, whenever practical, shall interchange or be replaced by military lubricants. All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Lubrication intervals shall be in accordance with SAE J752 as recommended by the component manufacturer. Manual fluid level indicators shall be in accordance with SAE J48 and J614. Grease lube fittings shall conform to SAE J534 and shall be of the threaded types. A lubrication chart(s) shall be furnished in accordance with SAE J753 and shall include the interchangeable military lubricants as well as the commercial SAE J754 lubricant designations.

3.13 Servicing and adjustment. The contractor shall service and adjust each vehicle for delivery in operational conditon for the user. The contractor shall use his commercial checklist and include at least the following:

- Adjustment of suspension.
- Adjustment of brake system.
- Electrical system continuity checks, voltage drop.
- Torque of wheel lug nuts.
- Inflation of all tires.
- Lubricant checks and lubrication of all components with approved products.

3.14 Technical publications. The technical publications shall be in accordance with SAE J920 except as otherwise specified (see 6.2).

3.15 Workmanship. Defective parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All workmanship on the semitrailers shall be in accordance with the engineering and production standards of the heavy semitrailer industry. The semitrailers shall withstand the operation specified herein without permanent deformation, breakage of connections, malfunction, or component intererence caused by incorrrect workmanship. All parts of the semitrailer before and after painting shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.15.1 Fabrication.

3.15.1.1 Aluminum and steel fabrication. Aluminum and steel shall provide original quality surface finish and shall be free from kinks and sharp bends. Aluminum and steel having eroded surface is not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be free of burrs, slag, and sharp edges. Precautions shall be taken to avoid overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

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3.15.1.2 Fiberglass fabrication. Fiberglass reinforced plastic (FRP) shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform FRP thickness. All edges and openings of FRP shall be completely sealed to prevent water intrusion. The FRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the FRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

3.15.2 Welders and welding.

3.15.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding ooperation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
Equal welder qualification program, subject to Government approval.

3.15.2.2 Metal welding. Welding shall be in accordance with the AWS and ASME codes, the American Welding Society's Welding Handbook, and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses. All welds made on high stress members shall be made from core or heavy-coated electrodes, gas shielded, or submerged arc. Electroslag and electrogas processes, when available, shall be included for use on heavy sections, 1 inch or more in thickness. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer, or each layer after the third pass with multipass welding process, shall be throroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. After welding, all weldments shall be free from slag (except mill scale), flux, weld spatter, and other factors which affect the qualifty of the weld. Except for vertical welding of the electroslag process, the work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Where different metals are welded to each other, filler metal having strength of the lower strength material shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification. Weld joints exceeding 50,000 psi shall be inspected for soundness.

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3.15.2.3 Aluminum welding. Aluminum welding shall be accommplished by the gas-metal-arc or the gas-tungsten-arc method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; hammered welds shall not be acceptable. Preheating for welding is permissible, providing the temperature does not exceed 400° F for a total time of 30 minutes. Aluminum weld processes requiring the use of welding flux shall not be used.

3.15.3 Brazing. Brazing of the aluminum and aluminum alloys shall be performed in accordance with the ASME code and other nationally recognized and accepted recommended procedures and standards of the aluminum manufacturing industry.

3.15.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Prevailing torque nuts are perferred in lieu of standard nuts and lockwashers. All capscrews, bolts, washers, and nuts shall be tight and correctly torqued. All fasteners and washers shall be finished with a corrosion preventative protective coating (including an eletrodeposited or mechanical plating or chemical conversion coating).

3.15.4.1 Critical bolted connections. Critical load-carrying bolted connecttions shall, in addition, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high-strength bolts with washers and nuts, hex or 12-point type. Bolts shall conform to IFI Standards 103 and 111, minimum SAE Grade 8, ASTM A449. Washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the components members shall be prevented. Mating nuts shall be of the prevailing torque type in accordance with IFI Standards 100, 101,107, and 108, minimum class C, unflanged, or class G flanged. Fasteners shall be lubricated prior to wrenching. To insure constant and correct torque performance, fastener installation accuracy shall be achieved by the turn-of-the-fastener method.

3.15.5 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burr removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.15.7 Forgings. Forged components shall be in accordance with the Forging Industry Association Handbook.

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3.15.8 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

3.15.9 Cleanliness of hydraulic and pneumatic systems. The hydraulic and pneumatic systems shall be purged to remove any foreign material after the appropriate assembly phase is completed. The sequence of purging shall be first through the lines and then second through both, the lines and the components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government vertification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to specification criteria has been demonstrated.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the referenced documents.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article semitrailer inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging for delivery (see 4.6).

4.3 First article semitrailer inspection. Prior to examination or tests, the semitrailer shall be lubricated with lubricants specified in 3.12, and tires shall be inflated to T&RA recommended pressure. The hydraulic system shall be filled with oil specified in 3.7. Prior to examination and test, the semitrailer shall be run-in in accordance with the contractor's standard procedure. No maintenance shall be performed on the semitrailer throughout the test specified in 4.5.2, except scheduled maintenance specified in the dump semitrailer manuals.

4.3.1 Examination. The semitrailer shall be examined for the defects marked "X" in column 1 of table IV. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the semitrailer shall be subjected to the test marked "X" in column 1 of table V. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each semitrailer shall be examined as specified in column 2 of table IV. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each semitrailer shall be tested as specified in column 2 of table V. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. Examination shall be in accordance with table IV.

TABLE IV. Examination Schedule.

First Article	Qual. Conf.	Defect	Requirement paragraph
1	2	3	4
X	–	101. Weights and dimensions not as specified.	3.1.1 & table I
X	–	102. Stress in any load carrying member exceeding 33 percent of the yield point of the material.	3.1.3
X	X	103. Truck-tractor semitrailer compatibility not as specified.	3.1.2
–	X	104. Electromagnetic interference of semitrailer with auxiliary engine not as specified or not identical with the first article model.	3.2.1
X	X	105. Cleaning treatment and painting not as specified.	3.2.2
X	X	106. Data plates not as specified.	3.2.2 thru 3.2.2.1.5
X	X	107. Vehicle marking not as specified.	3.2.3
X	X	108. Safety aspects not as specified.	3.2.4
X	–	109. Lifting and tiedown attachments not as specified, when required.	3.2.5 & 3.2.6
X	X	110. Dump body not as specified.	3.4 & 3.5
X	X	111. Hoist system not as specified.	3.6 & 3.8
X	X	112. Hydraulic system or components not as specified.	3.7
X	X	113. Hydraulic system lines or fittings not as specified.	3.7
X	X	114. Hydraulic cylinder not as specified	3.7
X	X	115. Semitrailer components not as specified.	3.9, table III

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TABLE IV. Examination Schedule (continued).

First Prod.	Qual. Conf.	Defect	Requirement paragraph
1	2	3	4
X	X	116. Upper fifth wheel and front supports not as specified.	3.9, table III
X	X	117. Suspension components not as specified, incorrect alinement, or inadequate rating.	3.9, table III
X	–	118. Loading of any tire exceeding highway or ML rating.	3.9, table III
X	X	119. Brake system not as specified.	3.9, table III
X	X	120. Electrical system not as specified.	3.9, table III
X	X	121. Lubrication system not as specified.	3.12
X	X	122. Servicing and adjustments incorrect, incomplete, or not as specified.	3.13
X	X	123. Workmanship not as specified.	3.15 thru 3.15.9
X	X	124. Leaks in air or hydraulic lines.	section 3
X	X	125. Seals do not prevent entrance of foreign material or do not prevent leak age of fluid or grease.	section 3
X	X	126. Assembly incorrect or not complete.	section 3
X	X	127. Parts or components missing or do not function.	section 3

4.5.1.1 Stress calculations. Stress calculations (see 3.1.3) for the load-carrying members of the semitrailer shall be made. The theoretical load for the stress calculations shall weight 3,000 pounds per cubic yard for aggregate and 425 pounds per cubic yard for refuse, as applicable, for a truck. The minimum payload shall be that specified in table I and the applicable specification sheet. Stress in any load-carrying member in excess of 33 percent of the yield point of the material shall constitute failure. Stress calculations for the upper fifth wheel plate assembly shall be as specified in 3.9, table III.

4.5.2 Tests. Tests shall be in accordance with table V and as specified herein. The contractor may use any truck-tractor for tests in 4.5.2.2 through 4.5.2.4 and 4.5.2.6, with the provision that the contractor's selected truck-tractor model has a published GCWR by its manufacturer equal to or in excess of the actual GCWR of the maximum test weight of the combination and conforms to the safety, performance, and dimensional compatibility requirements specified herein.

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TABLE V. Test Schedule.

First Prod.	Qual. Conf.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	–	Road test.	4.5.2.2	3.3 thru 3.3.3 & 3.9, table III
X	–	Brake test.	4.5.2.3	3.3.3 & 3.9, table III
–	X	Brake test.	4.5.2.3(c)	3.3.3
X	–	Dump semitrailer operation.	4.5.2.4	3.3 thru 3.3.2 & 3.4 & 3.5
X	–	Hydraulic system.	4.5.2.4	3.3 thru 3.4.2 & 3.6 & 3.7
–	X	Five mile performance and operation.	4.5.2.6	section 3
X	–	Lifting attachment.	4.5.2.7	3.2.5
X	–	Tiedown attachment.	4.5.2.8	3.2.6
X	–	Electromagnetic interference.	4.5.2.9	3.2.1

4.5.2.1 Test conditions. The test apparatus minimum standards of accuracy shall not exceed the following:

Type of Measurement	Percent of Measured Value
(a) Deceleration.	±2
(b) Force.	±2
(c) Linear dimension.	±1
(d) Speed.	±2
(e) Stress.	±2.5
(f) Weight.	±0.5

4.5.2.2 Road test. Unless otherwise specified (see 6.2), the following road test shall be performed. Couple the semitrailer to the truck-tractor and tow as specified in table VI for a total distance of at least 200 miles. Where table VI specifies that the semitrailer is loaded, the load shall be the payload specified in table I and on the applicable specification sheet.

TABLE VI. Road Test.

Course	Mileage and Speeds	Loading
Primary highway	50 mi. at variable speeds up to 55 mph	Loaded
Primary highway	50 mi. at variable speeds up to 55 mph	Empty
Unpaved roads	25 mi. at variable speeds up to 30 mph	Loaded
Unpaved roads	25 mi. at variable speeds up to 30 mph	Empty
Cross-country	25 mi. at variable speeds up to 20 mph	Loaded
Cross-country	25 mi. at variable speeds up to 20 mph	Empty

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During the road test and while towing rated payload, perform the following and record actual results:

(a) Tow the semitrailer on side slopes up to 10 percent, having the curb side downhill and having the road side downhill.

(b) Tow the semitrailer over 7-inch-high obstacles such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure the diagonals and parallels for evidence of misalinement. Questionable tire-to-surface contact shall be retested with single sheets of newspaper under the tires. Ability to remove the newspaper without tearing from under any tire shall be evidence of insufficient tire-to-surface contact.

(c) For tandem and multiple axle semitrailers only, weigh the weight imposed on each semitrailer axle (each axle on a separate scale) simultaneously and record the weight on each axle. Jack up each axle separately and return it to its scale, again recording the weight imposed on both the raised axle and those not raised. After each axle has been raised and all axle weights recorded, compare the suspension equalization differences between axles.

(d) Tow the semitrailer over standard, US Bureau of Public Roads V-ditches not less than 36 inches wide and 18 inches deep. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel for at least 10 times total.

(e) Measure and record the deviation or off tracking of semitrailer with respect to the prime mover, when the semitrailer is towed straight ahead on a paved surface.

(f) Jackknife the semitrailer at angles up to 90 degrees from straight ahead both to the left and to the right.

(g) Turn the prime mover, towing the semitrailer, in a suspension, pivoting sharp-turning circle for two complete turns and repeat in the opposite direction.

(h) Park the truck-tractor semitrailer combination, that is equipped with semitrailer supports, on a paved surface. Lower the supports to a height of 4 inches from the pavement. Uncouple the brake lines, setting the brakes, and unlock the tractor's fifth wheel jaws. With the tractor in low-low or in second-low gear, quickly engage the drive train and immediately accelerate the tractor out from under the front of the semitrailer, rapidly removing all support at the front of the semitrailer and allowing the supports to strike the paved surface in free fall. Repeat the supports impacts for a total of at least two free falls. Inspect both the supports and the semitrailer. Operate the supports from fully retracted to fully raised at least five times after the test.

(i) While dumping the payload, measure the semitrailer and all contractor furnished components in accordance with SAE J1077 to insure sound level compliance with both DOT Noise Emission Standards and DOL OSHA regulations on construction job site, based on a normal 8-hour shift per day and 20-miles to go for payload.

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Failure to complete the test, inability to operate and function as specified herein, or nonconformance to 3.3 thru 3.3.3 and 3.9 table III shall constitute failure of this test.

4.5.2.3 Brake test. Test the semitrailer for conformance to the DOT Standards and Regulations, the California Vehicle Code, and to the requirements specified herein. Tests shall be conducted with rated payload, unless different loading is specified.

(a) Tow the semitrailer over the test course and, when the semitrailer reaches the required speeds, conduct service, emergency, and parking braking ability tests in accordance with both the DOT and the California Vehicle Code procedures and requirements.
(b) With the semitrailer stopped on a paved surface, uncouple the truck-tractor brake lines from the semitrailer and bleed down all semitrailer air reservoirs. Determine by the drawbar pull method or on actual grades the maximum holding ability of the semitrailer emergency-parking brakes. Release the emergency parking brakes using only tools supplied with each semitrailer or truck-tractor and without resupply of compressed air.
(c) Test the semitrailer airbrake system as specified in TTMA RP No. 12 and RP No. 44, including anti-lock, when legally DOT required and furnished.

Failure to complete the test, inability to function and operate as specified, or nonconformance to DOT, the California Code, to 3.3 thru 3.3.3 and 3.9, table III, shall constitute failure of this test.

4.5.2.4 Dump semitrailer operation. With the semitrailer coupled to the truck-tractor, operate the semitrailer as follows:

(a) Perform dump cycles of loading, hauling and dumping of sand and aggregate, load to specified payload (refuse specials, substitute refuse and rubble).
(b) Tow the loaded and the empty semitrailer over a route subject to the approval of the contracting officer to and from the dump site.
(c) Dump the payload.
(d) Load, haul, dump, and return haul the semitrailer in operation (a) through (c) for not less than one 8-hour-work-shift.
(e) Load the semitrailer with crushed rock or stone, graded 3-inch-minus, to specified payload with a material free fall height of at least 3 feet above the body floor.
(f) Load with a scoop loader.
(g) Load from an overhead bin or hopper.

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(h) On a cushion base of earth or sand, except for refuse specials, other semitrailers load and dump a load of oversize boulders, shot rock, or concrete rubble having a largest diameter dimension of 5-feet or larger; dump with tailgate hinged at the bottom or tailgate removed.
(i) Dump payloads in a fixed location for stockpiling.
(j) Dump in a controlled uniform spreading lift of the payload while traveling forward.
(k) Dump the payload with the semitrailer on a side slope of at least the minimum slope shown in figure 2 (no wind conditions).
(l) Except for refuse specials, other semitrailers dump the hot mix payload into the hopper of a bituminous paver, of the wheel roller type or simulate dumping of hot mix by use of pea gravel.
(m) Demonstrate the dump body hoist system drift by insuring that the semitrailer is loaded with an evenly distributed load in the dump body to not less than the semitrailer's rated GVWR. Hydraulically raise the dump body from travel position to not less than 8-inches and not more than 18-inches, measured not more than 36-inches rearward from the front of the dump body. Place the dump control in the "Hold" position and place nonyielding, post type supports on the curb and road sides of the body exactly 2.0 inches lower than the hoisted, measurement point stated above. Observe the hoist system drift and measure and record the initial and final body position after a 15-minute time period.
(n) Check the hydraulic system throughout all testing. Install a gage into the pump suction line immediately adjacent to the pump inlet and install another gage into the pump outlet at the required pump test point. Monitor the gages periodically during the testing and record readings.

Failure to load, haul, and dump as specified herein or any evidence of instability, malfunction, permanent deformation, or breakage such as stress cracking of exterior paint, weld failure, bending of framing or reinforcing members, binding or leaking of the dumping mechanism, misalinement or hoist-system, drift in excess of the limit specified, touching or resting of the dump body on any support, or nonconformance to 3.3 through 3.13 shall constitute failure of this test.

4.5.2.4.1 Reliability and maintainability. When specified (see 6.2), the Government will perform the following two tests:

(a) Mean-time-between-failure. The semitrailer shall be tested in accordance with 4.5.2.4 for sufficient length of time to reach a decision to "accept" or "reject" in accordance with MIL-STD-781, test plan VI. A "failure" for the purpose of this test is defined as any random malfunction which causes degradation below the operating condition required to continue this test and cannot be remedied within 4 hours. Nonconformance to 3.3.5.1 shall constitute failure.

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(b) Maintainability. Maintenance operations required during reliability testing shall be accomplished in accordance with the maintenance ratio specified in 3.3.5.2. Nonconformance to 3.3.5.2 shall constitute failure of this test.

4.5.2.5 Prime mover test. When a prime mover is furnished with the dump semitrailer (see 3.1.2 and 6.2), inspect the prime mover in accordance with the applicable prime mover specification and this semitrailer specification.

4.5.2.6 Five mile production performance and operation test. Tow each production semitrailer, without load, a distance of not less than 5 miles over hard surfaced roads with the lights on. Operate the semitrailer hoist system to maximum dump two times. Maneuver and uncouple and park the semitrailer, including lowering of the front supports. Failure of the semitrailer to conform to all the requirements in section 3 shall constitute failure of this test.

4.5.2.7 Lifting attachment test. When lifting attachments are furnished, lift the semitrailer and hold in suspension for at least 3 minutes in its normal travel position with the loads imposed on the lifting attachments as specified in 3.2.5. Failure or permanent deformation of any attachment or any member of the semitrailer, or nonconformance to 3.2.5 shall constitute failure of this test.

4.5.2.8 Tiedown attachment test. When tiedown attachments are furnished, the static load as specified in 3.2.6 for tiedown attachments shall be applied to the semitrailer. Failure or permanent deformation of any attachment or any member of the semitrailer, or nonconformance to 3.2.6 shall constitute failure of this test.

4.5.2.9 Electromagnetic interference. The first article model when equipped with an auxiliary engine, shall be tested to determine conformance with 3.2.1. The contractor shall furnish the contracting officer the test report. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging for delivery. The preservation and packing shall be examined to determine compliance with the Quality Assurance Provisions of MIL-STD-281. The marking shall be examined to determine compliance with the referenced document. Any marking that is missing, illegible, incorrect, or incomplete shall be cause for rejection.

5. PACKAGING FOR DELIVERY

5.1 Vehicle manufacturer's processing. Unless otherwise specified herein, the semitrailers shall be processed for shipment from the manufacturer's plant to initial receiving activity in accordance with the manufacturer's standard commercial practice.

5.1.1 Marking. The statement "NOT PROCESSED FOR STORAGE" shall be stenciled on a plywood or particle board of appropriate length and width. Lettering shall not be less than 3/4-inch high and normally shall be visible at a distance of 20 feet. The board shall be wired or secured to the front of the semitrailer.

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5.2 Military processing. When specified (see 6.2 and 6.8), the semitrailers shall be processed in accordance with 5.2.1.

5.2.1 Preservation, packaging and marking. Each semitrailer shall be preserved, packaged, and marked in accordance with MIL-STD-281, as specified for (mobile) preservation and packaging. The level of preservation and packaging shall be level A, B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. Size 10 semitrailers are intended for use in hauling and dumping refuse. Sizes 12, 20, and 25 semitrailers are intended for use primarily in transporting and dumping loose material on earthmoving jobs. All semitrailers are intended to couple with and be towed by commercial truck-tractors including those conforming to KKK-T-2108, KKK-T-2109, and MIL-T-52901.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, numbers, and dates of this specification and specification sheets.
(b) Size of semitrailer and its specification sheets required (see 1.2).
(c) When a prime mover with each semitrailer is required and what, specifications, minimum size features, and options are required (see 3.1.2, 3.6.1, 3.7, 4.5.2.5, and 6.4).
(d) Time frame required for submission of first article semitrailer (see 3.1.4).
(e) When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.1.4).
(f) When electromagnetic interference is required (see 3.2.1).
(g) Cite the appropriate service and what treatment, painting, and data plates are required (see 3.2.2).
(h) When treatment and painting of the semitrailer in accordance with MIL-T-704 is required and, optionally, when single coat wax-based compounds are acceptable (see 3.2.2).
(i) Cite the appropriate service and what marking is required (see 3.2.3).
(j) When lifting attachments are required (see 3.2.5).
(k) When tiedown attachments are required (see 3.2.6).
(l) When a lightweight, aluminum semitrailer is required in lieu of high-tensile steel and the maximum weight limit for the aluminum semitrailer's curb weight (see 3.5).
(m) When a power actuated tailgate trip control system is required, and in addition, when it shall be operated only during a service brake application by the driver (see 3.5, (o) and (p)).

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(n) When the dump angle of the body may be less than the minimum angle specified or may be of the horizontal discharge type (see 3.6 and applicable specification sheet).
(o) When a self-contained type hoist system is required in lieu of the standard truck-tractor hydraulic hose type and if the self-contained type hoist system is required, whether the engine in 3.8 shall be required to be gasoline or diesel type [see 3.6.1 (b)].
(p) When truck-tractor hydraulic system components or details are required for contractor mounting on the user's vehicle, contractor furnished vehicle, dealer mounting, or user mounting (see 3.6.1 (a) and 6.4). Information as to truck-tractor/semitrailer hydraulic system connections and the specifications of the intended truck-tractor system should be given, if known, to insure compatibility (i.e., truck make, vehicle identification number (VIN), model, and year; transmission make and model).
(q) When an air suspension system is required (see table III, VMRS 16-008).
(r) When and which sizes and types of tires other than wide base single tires are required (see table III, VMRS 17-001).
(s) When a spare tire and wheel is required and if the spare is to be semitrailer mounted or supplied for motor pool tire shop storage (see table III, VMRS 17-001).
(t) When wheel chock blocks and either cradels or hangers are required (see table III, VMRS 53-004-013).
(u) When front stiff-leg supports, in lieu of gear box types are required (see table III, VMRS 77-002).
(v) When both l2-v. and 24-v. electrical systems are required for both commercial and military design truck-tractors (see 3.10).
(w) Technical publication requirements (see 3.14).
(x) When road testing is not required (see 4.5.2.2).
(y) When the Government will conduct the reliability and maintainability tests specified in 4.5.2.4.2.
(z) When military processing is required (see 5.2).
(aa) Level of preservation and packaging required (see 5.2.1).

6.3 First article semitrailer. Any changes or deviations of subsequent production semitrailers from the approved first article semitrailers will be subject to the approval of the contracting officer. Approval of the first article semitrailer will not relieve the contractor of his obigation to furnish semitrailers conforming to this specification.

6.4 Details of user's intended truck-tractor. Prior to use of this specification for procurement, the following data for the applicable towing truck-tractor should be obtained from the user:

(a) Detailed characteristics of hydraulic hose type hoist system of truck-tractor.
(b) Make, model, year and if available, vehicle identification number of truck-tractor.
(c) Make and model of transmission.

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(d) Make and model of lower fifth wheel. (NOTE: It is strongly recommended that users of truck-tractors with full oscillating fifth wheels consider replacement of those fifth wheels with fore and aft rocking fifth wheels to avoid any possibility of dangerous operation.)
(e) Type of cab seat (bench or individual types).
(f) Regular production options (RPO's) and special production options (SPO's) which could have an adverse influence on the PTO installation or the trailer operation.

Upon receipt of above information the applicable hoist actuation system may be selected in 3.5.1.

6.5 Manuals. All manuals for the semitrailer should be required to comply with SAE J920. The parts manual should be in accordance with the ATA Vehicle Maintenance Reporting Standards. In addition, one copy of the TTMA Semitrailer Maintenance Manual should be required to be furnished by the contractor with each semitrailer, to augment his own maintenance manual.

6.6 Warranty. The contracting officer should include a standard commercial warranty clause in the contract.

6.7 Commercial OEM literature. The contracting officer should forward the verification of ratings and commercial literature to the engineering offices of the procuring activity for evaluation (see 3.1).

6.8 Military processing. Military processing should be specified when the dump semitrailers are destined for overseas shipment or when the anticipated storage time within the continental United States (CONUS) will exceed 1 year.

Custodian:
Army – ME

Preparing activity:
Army – ME

Review activity:
Army – AT

Project 2330-0745

User activities:
Army – CE
Navy – MC, YD

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